

16-Bit ALU Assignment Report

Mohamed Abdelsalam

July 8, 2026

1 Synthesis Diagram

The following figure shows the synthesis diagram of the 16-Bit ALU design provided by Intel/Quartus or the used synthesis tool.

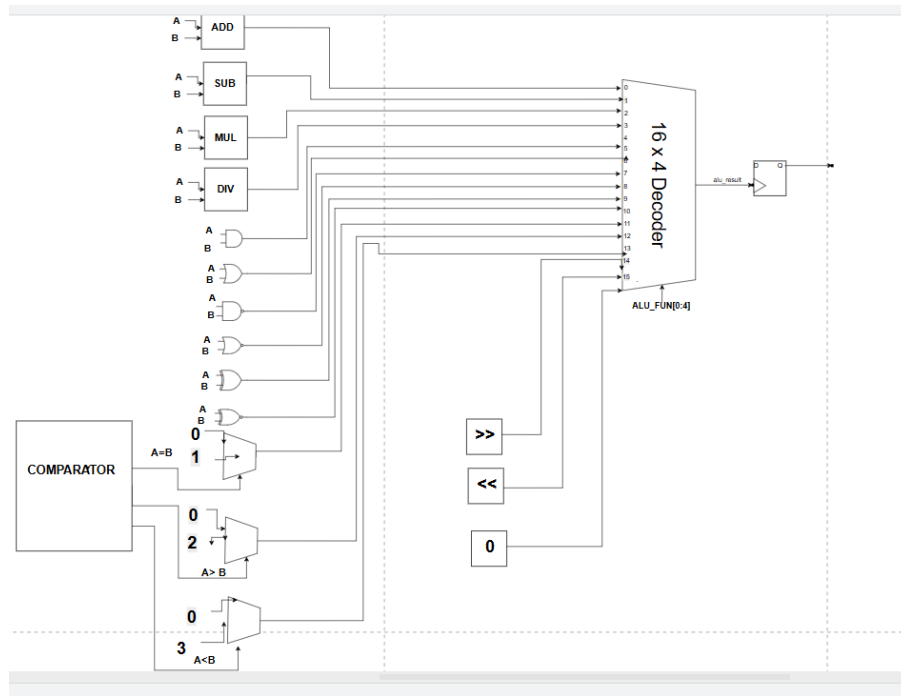


Figure 1: 16-Bit ALU Synthesis Diagram

2 Simulation Waveforms

The simulation waveforms demonstrate the functional correctness of the ALU operations over the 16 test cases. The waveforms are divided into logic, shift, arithmetic, compare, and carry flag groups.

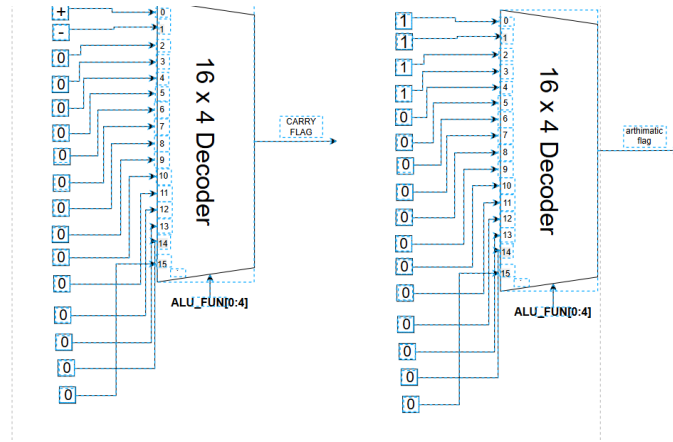


Figure 2: Carry and Arithmetic Flags Simulation Waveforms

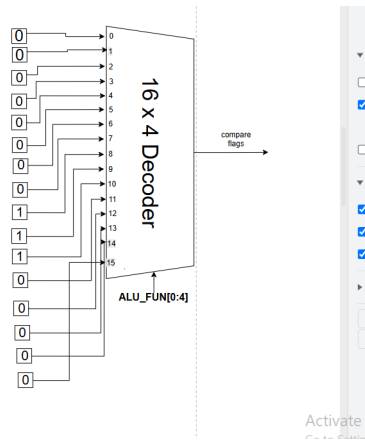


Figure 3: Compare Flags Simulation Waveforms

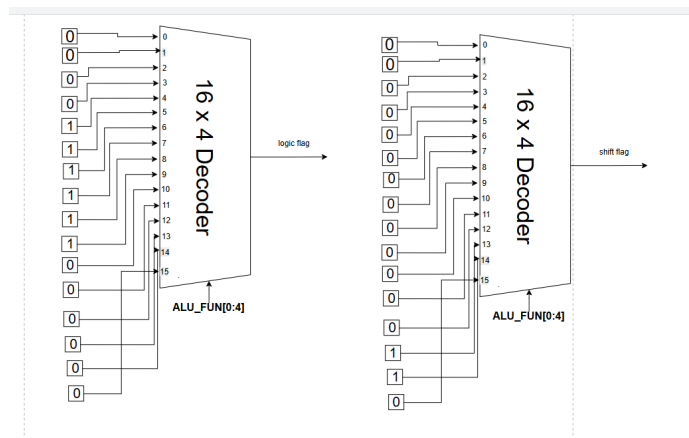


Figure 4: Shift and Logic Flags Simulation Waveforms

3 Simulation Logs

The output log from ModelSim verifies the numerical results of each ALU operation implemented in the testbench. All test cases output the corresponding ALU_FUN status, Flag results, and the output Data.

```
Reading C:/intelFPGA/18.0/modelsim_ase/tcl/vsim/pref.tcl
```

```
# 10.5b
```

```
# vsim -c tb_ALU_16B -do "run -all; quit"
# Start time: 23:22:28 on Jul 08,2026
# Loading work.tb_ALU_16B
# Loading work.ALU_16B
# run -all
# ==== 16-bit ALU Testbench ====
# Clock: 100 KHz (period = 10 us)
# -----
# TC1 | FUN=0000 | A=100 B=200 | OUT=300 | Cy=0 Ar=1 Lo=0 CM=0 Sh=0
# TC2 | FUN=0000 | A=65535 B=1 | OUT=0 | Cy=1 Ar=1 Lo=0 CM=0 Sh=0
# TC3 | FUN=0001 | A=500 B=200 | OUT=300 | Cy=0 Ar=1 Lo=0 CM=0 Sh=0
# TC4 | FUN=0001 | A=10 B=20 | OUT=65526 | Cy=1 Ar=1 Lo=0 CM=0 Sh=0
# TC5 | FUN=0010 | A=15 B=4 | OUT=60 | Cy=0 Ar=1 Lo=0 CM=0 Sh=0
# TC6 | FUN=0011 | A=100 B=4 | OUT=25 | Cy=0 Ar=1 Lo=0 CM=0 Sh=0
# TC7 | FUN=0100 | A=65280 B=4080 | OUT=3840 | Cy=0 Ar=0 Lo=1 CM=0 Sh=0
# TC8 | FUN=0110 | A=65280 B=4080 | OUT=61695 | Cy=0 Ar=0 Lo=1 CM=0 Sh=0
# TC9 | FUN=0101 | A=43605 B=21930 | OUT=65535 | Cy=0 Ar=0 Lo=1 CM=0 Sh=0
# TC10 | FUN=0111 | A=43605 B=21930 | OUT=0 | Cy=0 Ar=0 Lo=1 CM=0 Sh=0
# TC11 | FUN=1010 | A=42 B=42 | OUT=1 | Cy=0 Ar=0 Lo=0 CM=1 Sh=0
# TC12 | FUN=1011 | A=100 B=50 | OUT=2 | Cy=0 Ar=0 Lo=0 CM=1 Sh=0
# TC13 | FUN=1100 | A=5 B=99 | OUT=3 | Cy=0 Ar=0 Lo=0 CM=1 Sh=0
# TC14 | FUN=1101 | A=43690 B=0 | OUT=21845 | Cy=0 Ar=0 Lo=0 CM=0 Sh=1
# TC15 | FUN=1110 | A=21845 B=0 | OUT=43690 | Cy=0 Ar=0 Lo=0 CM=0 Sh=1
# TC16 | FUN=1111 | A=1234 B=5678 | OUT=0 | Cy=0 Ar=0 Lo=0 CM=0 Sh=0
# -----
# Simulation complete.
# ** Note: $finish : tb_ALU_16B.v(109)
# Time: 166 us Iteration: 0 Instance: /tb_ALU_16B
# End time: 23:22:28 on Jul 08,2026, Elapsed time: 0:00:00
# Errors: 0, Warnings: 0
```